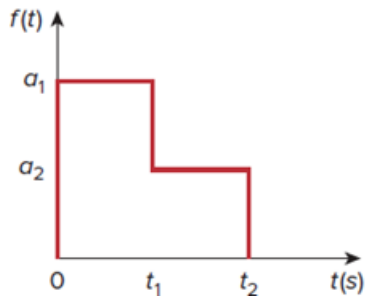


Problem

Using Fig. 15.29, design a problem to help other students better understand the Laplace transform of a simple, non-periodic waveshape.

Figure 15.29:



Step-by-step solution

Step 1 of 2

Refer to Figure 15.29 in the text book.

Determine the Laplace transform of $f(t)$.

$$\begin{aligned}
 F(s) &= \int_{-\infty}^{\infty} f(t) e^{-st} dt \\
 &= \int_0^{t_1} a_1 e^{-st} dt + \int_{t_1}^{t_2} a_2 e^{-st} dt \\
 &= a_1 \left(\frac{e^{-st}}{-s} \right) \Big|_0^{t_1} + a_2 \left(\frac{e^{-st}}{-s} \right) \Big|_{t_1}^{t_2} \\
 &= a_1 \left(\frac{1 - e^{-st_1}}{s} \right) + a_2 \left(\frac{e^{-st_1} - e^{-st_2}}{s} \right)
 \end{aligned}$$

Step 2 of 2

Simplify further.

$$\begin{aligned}
 F(s) &= \frac{a_1 (1 - e^{-st_1}) + a_2 (e^{-st_1} - e^{-st_2})}{s} \\
 &= \frac{a_1 + (a_2 - a_1) e^{-st_1} - a_2 e^{-st_2}}{s}
 \end{aligned}$$

Thus, the Laplace transform of $f(t)$ is $\boxed{\frac{a_1 + (a_2 - a_1) e^{-st_1} - a_2 e^{-st_2}}{s}}$.